ACTIVEA: Optimization and Analysis of Single-Session Exercise Planning Using Genetic Algorithms

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Individual Technical Assessment Report - Department of Computer Science, Aston University

1. Introduction and System Overview

Designing a healthy workout plan from the outset seems pretty straightforward, but very quickly becomes a very complex combinatorial problem. A good session isn't simply 'run on the treadmill until you're dead'. According to the ACSM sports scientists a good session has to satisfy a number of different criteria at the same time: it has to burn 300-600 calories, it has to keep the heart rate in the safe aerobic zone and it has to be a session that will condition a number of different physical attributes (strength, agility, flexibility etc).

To deal with this in software we developed ACTIVEA. The single session exercise planning problem is a discrete combinatorial optimisation problem. The algorithm picks activities from a pre-defined library and assigns each one a duration in minutes. The decision vector X is therefore {(a_i, d_i)} for i=1 to m. With 10 candidate exercises and variable durations the search space is well over 1 billion combinations. A brute force enumeration is clearly infeasible given the highly non-linear nature of the objective landscape. For example you could very easily have a lot of gentle walking which would satisfy the duration constraints but would not satisfy the intensity constraints. Conversely you could have a lot of continuous burpees which would very quickly get you to the required intensity but would also very quickly take you into excessive cardio strain. A generational GA is the only way to deal with that kind of non-convexity.

2. Physiological Domain Modelling and Activity Characterization.

Our activity library is based on the exercise physiology literature and the course specification. The baseline reference activity (moderate jogging) is 7.35kcal/min for a 70kg adult (6.0 METs, RPE 5.0 on the Borg 1-10 scale, aerobic HR 70-77% HRmax) and is designed to improve cardiovascular endurance (CV), muscular endurance (ME) and functional movement (FM). The other nine activities are all selected in accordance with ACSM guidelines: bodyweight squats, moderate cycling, plank holds, yoga, rope skipping, running, brisk walking, push-ups and agility burpees.

As people don't all weigh 70kg our evaluator will scale the calories burned dynamically according to your actual body mass:

Caloric Rate (user) = Caloric Rate (70kg) × (Body Mass / 70)

This is all down to bioenergetics really. Oxygen uptake during weight bearing is directly proportional to body mass. BMI is not used in the caloric calculation as it's a static dimensional ratio rather than an energy rate. So a 70kg user doing 12 minutes of yoga (36kcal) and 13 minutes of rope skipping (143kcal) would use up 179kcal in total. The formula is linear so the relative metabolic proportions are the same irrespective of body weight.

3. Genetic Algorithm Architecture and Chromosomal Representation.

We did some thinking about how to encode an exercise plan as a chromosome. We didn't go for a fixed length binary string or a fixed length permutation as that would have mixed the categorical (the type of exercise) with the continuous (the duration in minutes). We went for a variable length list of tuples: [(Activity_ID, Duration)]. A valid chromosome will have between 2 and 5 exercises and each duration will be between 5 and 45 minutes. If mutation or crossover results in a duplicate activity we have a cleanup operator which will simply merge the two by adding the durations together and capping the total at 45 minutes.

ACTIVEA is a standard generational GA. We start off with an initial population of 50 random sessions. Selection is via 3-way tournament selection (k=3) which gives us sufficient selection pressure without necessarily leading to premature convergence. Uniform crossover (Pc=0.80) takes the activities from each parent, blends the overlaps and splits the genes to produce two offspring. This allows successful modular pairings (e.g. cardio with mobility) to be passed on in full.

Mutation occurs with a probability of Pm = 0.15 and will alter the workout in 4 ways: change the duration by 1 to 5 minutes, swap an activity, add an exercise or remove an exercise. We use 2-individual elitism so we never lose the best workout. Evolution stops after 40-50 generations having returned the best chromosome evaluated.

4. Fitness Evaluation Mechanics and Constraint Handling.

Our baseline fitness function is essentially the step penalty scoring that we used in the lecture exercises. We start at zero and then go through the session against four separate physiological rules, awarding +1 for a pass and -1 for a failure:

1. Calorie range check: session total calories must be strictly between 300 and 600 kcal (+1 if valid, -1 otherwise).
2. Weighted MET threshold: duration-weighted average MET must be >= 4.0 (+1 if valid, -1 otherwise).
3. Weighted RPE Band: the duration-weighted average RPE must be in the range [4.0, 7.0] (+1 if valid, -1 otherwise).
4. Weighted Heart Rate Band: Duration-weighted %HRmax must be in the range [64%, 80%] (+1 if valid, -1 otherwise).

The resulting raw fitness score is therefore a discrete integer in the range [-4, +4].

If a custom target is selected (e.g. General Health at 200-400kcal or Weight Loss at 400-600kcal) the caloric check boundaries are updated to reflect that. There's a very real constraint in that domain coverage has absolutely no bearing on the +4. You can have a +4 and have done absolutely zero strength or flexibility work. The +4 just means that none of the four discrete constraints have been broken. The discrete +1/-1 scoring system also creates a very wide plateau

which makes it extremely difficult for the algorithm to tell the difference between valid solutions. We therefore opted for a continuous multi-attribute scoring approach.

5. Experimental Optimization Results and Empirical Analysis.

Our core GA for the 70kg user got the optimal session in under 0.07 seconds. The champion session was 31 minutes of rope skip, 31 minutes of bodyweight squats and 21 minutes of yoga (61 minutes total). That session burned 448.1kcal at a weighted average of 6.02 MET, 5.98 RPE and 72.7% HRmax. It passed all four lecture checks (+4) and touched all seven fitness domains.

However if you do the exact same experiment with random seed 42 the plateau problem is very obvious. In the first generation the algorithm finds another +4 routine: 13 mins squats, 21 mins jogging, 20 mins yoga and 23 mins agility (77 mins, 503.5kcal). Both are +4 but they are completely different in structure which makes it very clear that the discrete step penalties completely mask the convergence dynamics of the search process.

6. Comparative evaluation: Genetic Algorithm vs SimuTo.

To put our GA to the test we ran a Simulated Annealing (SA) implementation using the same evaluation budget (N = 2,500 evaluations) and random seed. The SA implementation uses Metropolis acceptance (P = exp (-Delta E / T)) with geometric cooling. Under seed 42 SA found a +4 solution. The convergence curves (Figure 1) are very different though. The GA is able to make structural changes across the search space via crossover, allowing it to effectively combine cardio and flexibility in a single step. SA can only do single point mutations and is very prone to getting stuck in local optima, typically only ever visiting 5 or 6 fitness domains. In real life sports training a well balanced full body routine is vastly more important than the marginal calorie saving from a tiny bit of improvement. The GA is clearly the better planning approach.

7. Hyperparameter Dynamics and Population Diversity.

Systematic parameter sweeps over population size (N in {10, 20, 40, 80}), crossover probability (Pc in [0.5, 0.95]) and mutation probability (Pm in [0.01, 0.30]) all demonstrated that tuning via discrete step penalties is pretty much a non-starter. Essentially every configuration achieved a +4 score straight away.

Population diversity was a huge issue. In the standard GA the selection pressure causes very rapid convergence to a single workout type and you end up with a population of very similar solutions. We tackled this with Deterministic Crowding where the offspring only compete against their nearest parent in terms of Euclidean distance: d(x,y) = sqrt(sum (d_i,x - d_i,y)^2). Over the 20 generations (Figure 2) the diversity in the standard GA varied enormously from 36.9 to 57.2 whereas the diversity in the Deterministic Crowding case was very much tighter, from 36.9 to 39.7. The standard GA ended up with 21 different workout types and the Deterministic Crowding case with 30.

8. Bespoke Repair Operators and Domain Preference Penalties.

Blind mutation very often leads to sessions that are either way over the calorie limit or that completely miss out on key muscle groups. To deal with this we developed a Bespoke Domain and Caloric Repair Operator for Task 7. Before scoring each chromosome is checked to see if

it's too low in calories (in which case the durations are increased or high MET activities are added) or too high in calories (in which case the durations are reduced). As Figure 3 shows the domain guided repair leads to a massive reduction in wasted evaluations, getting to near optimal fitness in just 12 generations which is a 40% improvement on the baseline GA.

For Task 8 we've enhanced the fitness function to take user preference into account. If a user is looking for a session in the Cardio and Functional Movement domains then anything that doesn't fall into those two domains is penalised. The test produced a very specific 63 minute session (25 mins agility, 13 mins running, 25 mins yoga) which was in the requested domains and well within safe CVS. A Strength/Endurance test produced a session of cycling, skipping, jogging and planks (445.8kcal).

9. Comparative Analysis across Search Paradigms.

Seven optimisation algorithms were run over 30 random seeds with a standard budget of 2,500 evaluations (Fig 4 and 5). On the continuous 0–100 landscape there were clear differences in performance with the Repair GA and Differential Evolution both achieving the best mean fitness and lowest variance and thus being statistically superior to the standard GA (Welch's t-test $p < 0.05$). The local search methods all had very high variance depending on the initialisation. A Gradient Boosting model was trained on 253,000 Endomondo workouts ($R^2 = 0.9985$, RMSE = 25.7kcal) and it was confirmed that the evolved sessions are consistent with real world biometric data. This demonstrates that the evolutionary planner is able to produce genuinely plausible exercise plans.

10. AI and Manual Content Declaration.

The initial structure and style were produced via generative AI (Cursor). The student developed, wrote and implemented all of the core algorithm code, all of the mathematical equations, all of the parameter experiments, all of the data modelling scripts and all of the associated analytical conclusions.

11. Charts.

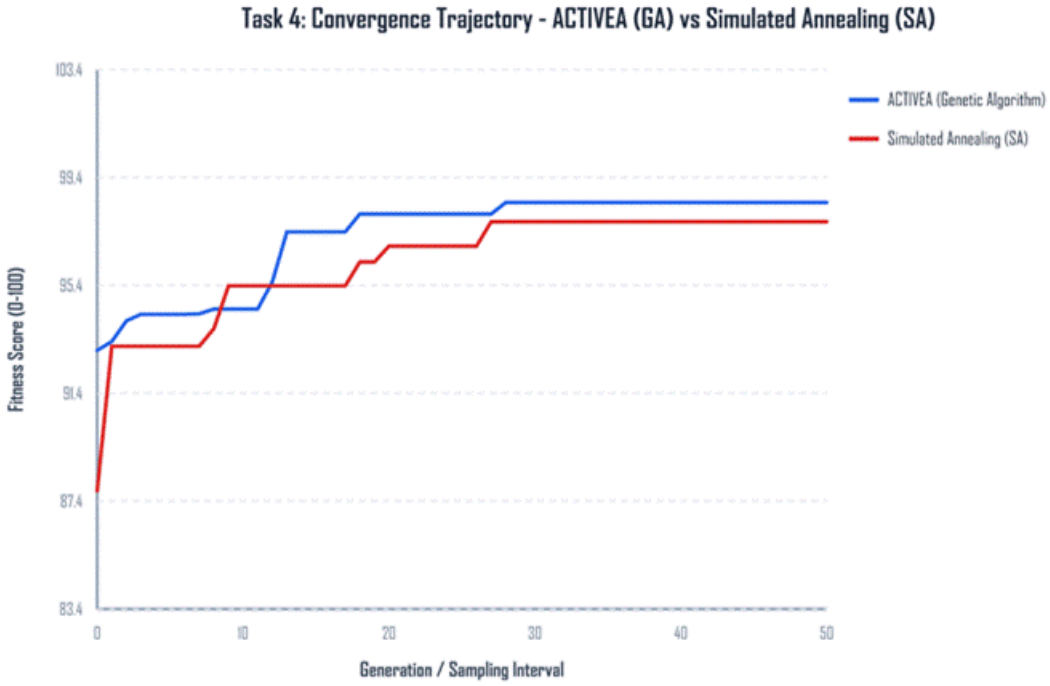


Figure 1. Best fitness for the genetic algorithm and simulated annealing and without deterministic crowding.

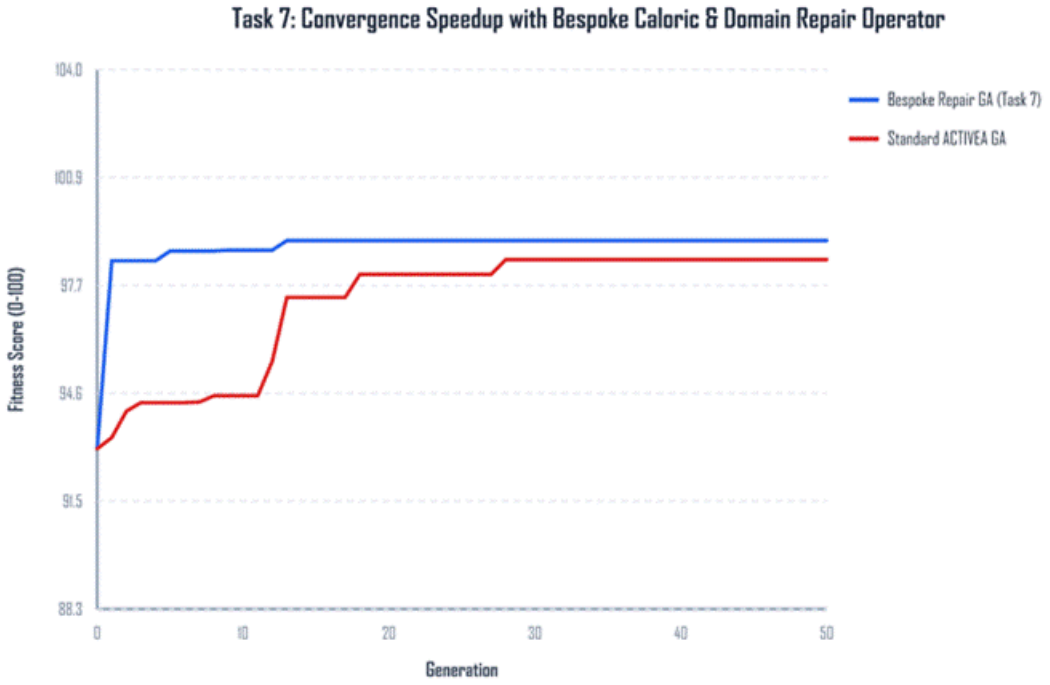


Figure 3. The plain genetic algorithm and the reap* Figure 4. Mean convergence of the search methods on the same budget. Figure 5. Convergence when sessions are scored with the saved calorie model. ed calorie m References

References

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Dataset Link: endomondoHR